C PROGRAMMING PROJECT PRESENTATION

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Project Title: Mini Attendance Tracker

Course: C Programming Lab

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Department: Computer Science

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INTRODUCTION

Overview of the Project and Its Purpose

The Mini Attendance Tracker is a simple console-based application designed to help teachers or class representatives efficiently record and manage student attendance during a session. The system allows users to set the class size, mark all students as present or absent, update attendance for individual roll numbers, and view summaries of present and absent students. Its purpose is to provide a quick and organized way to handle attendance data without needing manual paperwork.

Problem Background and Motivation

In many educational institutions, attendance is still tracked manually using registers or spreadsheets, which can be time-consuming and prone to human errors. Managing large class sizes becomes difficult, especially when attendance must be taken frequently. This project was motivated by the need for a lightweight, user-friendly, and efficient tool that can automate the attendance marking process within a single session, saving time and effort for both students and teachers.

Significance of the Project in Solving Real-World Issues

The Mini Attendance Tracker addresses common challenges in traditional attendance management systems. It ensures accuracy, speed, and simplicity, reducing the likelihood of missing or duplicate entries. By providing instant summaries and the ability to update records easily, it enhances classroom organization and productivity.

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OBJECTIVE

Goal of this project:

The project aims to create a simple yet effective system that can record, update, and display student attendance automatically during a session.

To apply core programming logic and modular design

The project focuses on using key C programming concepts such as loops, arrays, functions, and conditional statements to build a structured and modular application.

To demonstrate problem-solving and analytical skills

Through designing and debugging the program, the project helps in enhancing logical thinking, algorithmic approach, and the ability to translate real-world problems into working code.

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PROBLEM STATEMENT

Manual attendance tracking is slow, repetitive, and error-prone.

This project aims to automate the process for small classroom sessions by providing a simple, console-based interface that:

> Records attendance efficiently.

> Ensures accuracy through validation.

> Displays quick summaries and lists.

> All without the need for external files or databases.

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SYSTEM REQUIREMENTS

Hardware Requirements

Processor: Intel Core i3 or higher

RAM: Minimum 4 GB

Storage: At least 500 MB of free disk space

Software Requirements

Operating System: Windows or Linux

Compiler: GCC / Turbo C / Code::Blocks / Dev C++

Editor: Visual Studio Code / Notepad++

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SYSTEM DESIGN

The Mini Attendance Tracker follows a modular and menu-driven architecture. Each user operation is handled by a dedicated function to improve clarity and maintainability.

1. Structure and Workflow of the System

Main Workflow

Start → Initialise all roll statuses to '-'.

Display Menu → Show user options (1–8).

User Input → Read and validate choice.

Process Choice:

Set Class Size → Define number of students (N).
Mark All → Record attendance for all rolls (P/A).

Update Roll → Change one student's status.

Show Summary → Count present/absent & calculate percentage.

Show Present/Absent Lists → Display respective rolls.

Reset Session → Clear all statuses to '-'.

Exit → End program.

Repeat Menu until user selects Exit.

2. Flowchart (Logic Flow)

START

↓

Initialise status[]='-

↓

Display Main Menu (1–8)

↓

Read User Choice

↓

Perform Selected Action
(Set Size / Mark / etc.)

↓

Choice == 8 ? (Exit)

yes ↓ no ↓

Exit ← **Show Menu**

3. Algorithm Outline

Step 1: Start
Step 2: Display main menu
Step 3: Get user choice
Step 4: Uses switchcase,
If choice = 1 → Set/Change class size
If choice = 2 → Mark attendance (P/A) for all roll
If choice = 3 → Update one roll
If choice = 4 → Show summary
If choice = 5 → Show present list
If choice = 6 → Show absent list
If choice = 7 → Reset session
If choice = 8 → Exit
Step 5: Repeat until user selects Exit
Step 6: Stop

4. *Block Diagram (System Modules)*

1. Input Module
   → Accepts class size, roll updates

2. Processing Module
   → Marks attendance, updates records

3. Display Module
   → Shows summary, present & absent

4. Control Module
   → Handles menu navigation & choices

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MODULE DESCRIPTION

Input Module – Takes user input for class size, attendance (P/A), and roll updates. Performs validation to ensure correctness.

Processing Module – Handles logic for marking attendance, updating rolls, counting present/absent, and calculating percentage.

Output Module – Displays the attendance summary, present and absent lists, and confirmation messages to the user.

Control Module – Manages the main menu and coordinates function calls for each selected operation.

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ALGORITHM / FLOWCHART

Step 1: Start the program
Step 2: Initialise necessary variables and arrays for storing attendance data
Step 3: Display the main menu with available options
Step 4: Get user input (choice)
Step 5:
    If choice = 1 → Set/Change class size
    If choice = 2 → Mark all present
    If choice = 3 → Update one roll number
    If choice = 4 → Show summary
    If choice = 5 → Show present list
    If choice = 6 → Show absent list
    If choice = 7 → Reset session
    If choice = 8 → Exit
Step 6: Perform the selected operation
Step 7: Display the result or updated status to the user
Step 8: Repeat steps 3–7 until the user exits
Step 9: Stop the program

Flowchart

Start

|

Initialise Data

|

Get User Input

|

Process Data

|

Display Output

|

Stop

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IMPLEMENTATION

Techniques Used:

Loops and Conditional Statements – Used for menu repetition, roll marking, and decision-making (for, while, if-else, switch).

Functions – Each operation (set class size, mark all, update, summary, etc.) is written as a separate function.

Arrays – A 1D array status[MAX] is used to store attendance for each student.

Input Validation – Ensures only valid inputs (P, A, or valid roll numbers) are accepted.

Formatted Output – printf() , scanf() is used.

INPUT AND OUTPUT SCREEN

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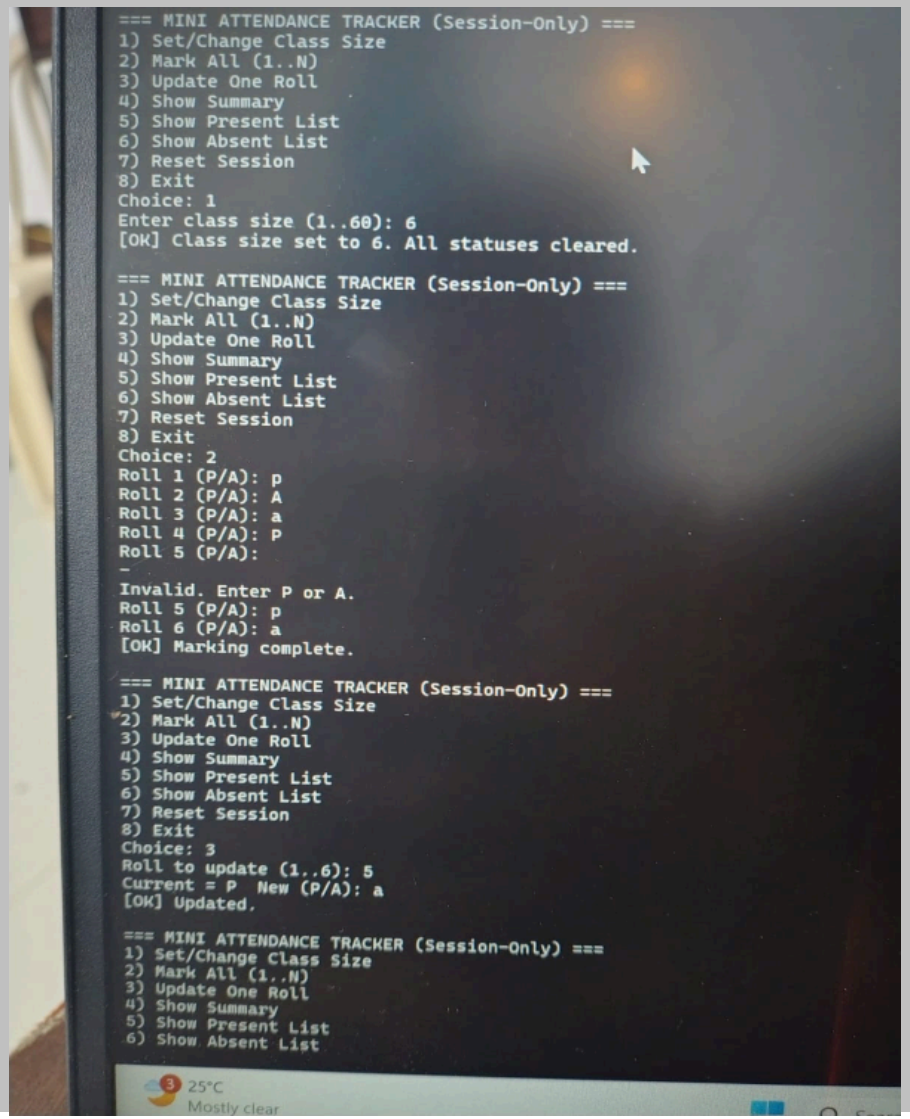

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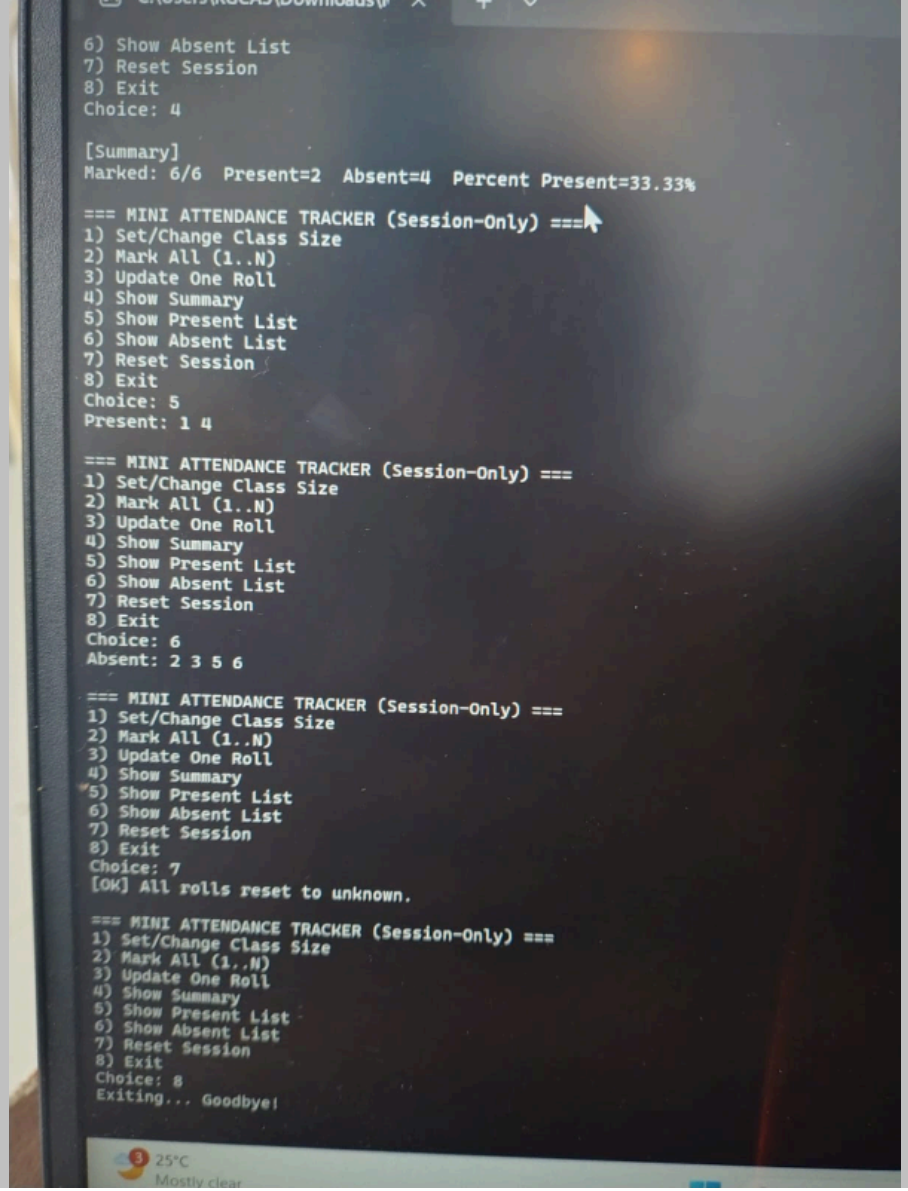

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RESULTS & DISCUSSION

The Mini Attendance Tracker performed successfully according to the project objectives. The program accurately recorded and displayed attendance for any class size between 1 and 60 students. It efficiently generated summaries showing the number of students present, absent, and the overall attendance percentage.

The system's outputs matched the expected results across various test cases — including all-present, all-absent, and mixed attendance scenarios. Input validation prevented crashes and ensured consistent accuracy. The use of arrays and functions contributed to fast execution and a clean, modular structure.

During implementation, challenges such as handling invalid inputs and ensuring correct percentage calculations for partially marked sessions were resolved using robust input validation and conditional logic. Overall, the project demonstrates a reliable, well-structured, and efficient application of core C programming concepts to a practical problem.

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CONCLUSION & FUTURE ENHANCEMENT

Conclusion:

Project achieved its main goal of building a session-based attendance tracker.
Strengthened understanding of C concepts, logic flow, and modular design .
Demonstrated practical application of arrays, loops, and functions.

Future Enhancements:

GUI support for better interaction can be added, Integrate database for record storage, using extend features for scalability and automation can also be added for the better future.

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REFERENCES / ACKNOWLEDGEMENT

Reference

- Let Us C – Yashavant Kanetka
- C Programming – E. Balagurusamy
- Online resources and documentation

Acknowledgement:

I sincerely express my gratitude to my mentor for their valuable guidance and support. Special thanks to my department for providing resources and I also appreciate my team members for their cooperation and contribution throughout the project.

THANK YOU

Every roll counts — track it right, code it light!

Questions:
Why is an array used to store attendance instead of another data type?
What does toupper() do in this code?
What happens if the user enters a roll number outside the valid range?